Krishna Krishna

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.60
Modern Academy	Class XII (ISC)	2024	97.60%
Modern Academy	Class X (ICSE)	2022	96.00%

SCHOLASTIC ACHIEVEMENTS

Solved 600+ data structures & algorithms problems across various online coding platforms (Codeforces, LeetCode, GFG)

PROJECTS

SciFact Scientific Claim Verification | Python, Hugging Face Transformers DeBERTa-v3-base, BM25, FAISS

- **Classified scientific claims** as Support, Contradict, or Not Enough Info with a **two-stage retrieve-then-verify pipeline** built from scratch, tested against a 5000-abstract corpus using a custom diagnostic suite designed to probe polarity sensitivity and negation handling.
- **Raised Contradict F1** from 0.64 to 0.75 and macro F1 to 0.82 by diagnosing a polarity-insensitivity failure in the fine-tuned DeBERTa-v3 verifier, then building **ClaimNegator, a self-built rule-based negation augmentation tool**, and retraining the verifier end to end on the augmented claim pairs.
- Improved retrieval recall at top 3 results by 8.5 points, from 70.2 percent with hybrid RRF fusion to 78.7 percent with BM25 alone, by running a full retrieval ablation across dense, BM25, and hybrid methods at multiple cutoffs on a held-out set of 2655 claims and selecting **BM25 as the production retriever.**
- Exposed a **26 point accuracy gap** between oracle-evidence and full-pipeline conditions, from 90 percent down to 65 percent, by building a two-stage evaluation harness that cleanly isolates retrieval quality from verifier quality across the entire pipeline.

IITH Freelance Marketplace | MongoDB, Express.js, React, Node.js (MERN)

- Built a **full-stack freelance marketplace** scoped to the IITH community, structuring transactions as a **job, bid, order, checkpoint, and review** flow instead of a single upfront payment with no verification step
- Designed a **two-stage checkpoint** workflow per order where checkpoint rejection cancels the order and reopens the job for rebidding, and bid acceptance auto-rejects all competing bids while transitioning job, bid, and order state together
- **Eliminated a race condition** allowing two buyers to simultaneously accept the same bid, by replacing a read-then-write status check with an atomic MongoDB findOneAndUpdate guard, guaranteeing exactly one winner under concurrent requests
- **Prevented orphaned orders** and permanently-stuck job states from partial-write failures, by wrapping multi-document writes (bid/job/order/checkpoint) in MongoDB **transactions** and validating **rollback** by deliberately injecting mid-transaction faults

Student Stress Predictor | Python, Flask, scikit-learn, XGBoost, MLflow

- Built an **end-to-end ML pipeline** predicting student stress level (High/Moderate/Low) from lifestyle habits (study/sleep/social/ activity hours, GPA), structured as modular ingestion, transformation, training, and inference stages
- Needed to identify the best classifier without manual reruns; trained and benchmarked **7 classifiers (Logistic Regression, Random Forest, Gradient Boosting, XGBoost, AdaBoost, KNN, Decision Tree)** under a single MLflow experiment, tracking accuracy, precision, recall, and F1 per run.

SKILLS

Languages: Java, Python, JavaScript, HTML, CSS, SQL, MATLAB

Machine Learning: NLP, Transformers, BM25, FAISS, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Fundamentals: Data Structures & Algorithms, Operating Systems (OS), Database Management Systems (DBMS), Object Oriented Programming (OOPs)

Web Development: React.js, Express.js, Node.js, MongoDB

Tools: Git, GitHub, Postman, MLflow

Soft Skills: Problem Solving, Analytical Thinking, Communication, Team Collaboration, Coordination, Time Management

EXTRACURRICULAR

Volunteer, Bisnouli Sarvodaya Gramodyog Sewa Sansthan (NGO)

- Conducted field-based need-assessment and community outreach for a women's vocational training program in Khora Colony, engaging local stakeholders, mobilizing beneficiaries, and teaching basic English and computer literacy sessions.

RELEVANT COURSES

Computer Science: Introduction to Programming

Mathematics: Introduction to Probability, Introduction to Statistics, Elementary Linear Algebra, Calculus-I, Calculus-II, Differential Equations, Transform Techniques, Complex Variables